

Kirin P. Danek

✉ kd9132@alumni.princeton.edu 🌐 kirin2d2.github.io 📷 [Kirin2D2](#) in [kirindanek](#)

Research Interests: I develop computer vision and machine learning systems that are **fair, transparent, and interpretable**. My work spans concept-based interpretability and model compression, mechanistic analysis of bias in LLMs, and human-centered evaluation of AI systems, drawing on human-computer interaction and cognitive science.

Education

Princeton University, Princeton, NJ

May 2026

B.S.E. in Computer Science

Departmental GPA: 4.0; **Overall GPA:** 3.82 (/4.0)

Minor in Statistics and Machine Learning

Research

Princeton University

Dec 2024 – Present

Concept-Aware Neural Network Pruning

Advisor: Dr. Vikram Ramaswamy

- Developed Concept-Aware Network Pruning (CNP), a structured pruning framework that identifies semantically meaningful concepts within a CNN's decision strategy and removes components tied to spurious correlations, without requiring spurious-concept labels at pruning time.
- CNP maintains out-of-distribution accuracy at up to $4\times$ the compression rate of prior pruning methods, improving group robustness while preserving in-distribution accuracy.
- Poster/extended abstract presented at the XAI4CV Workshop at CVPR 2026.

Princeton University

Feb 2026 – Present

Mechanistic Analysis of Racial Dialect in LLMs

Advisors: Dr. Lydia Liu, Dr. Tri Dao

- Identify minimal computational subgraphs (circuits) in GPT-2 for standard interpretability tasks (Indirect Object Identification, Gendered Pronoun) under African American English (AAE) vs. Standard American English (SAE) prompts via edge pruning.
- Find that the AAE circuit overlaps only 25% with the SAE circuit — dialect is a larger driver of circuit divergence than prompting strategy.
- Audit differences in ChatGPT-5.3 citation hallucinations conditioned on AAE vs. SAE user dialect (results pending).

Princeton University

Sep 2025 – Dec 2025

CogniSplit: LLM-Mediated Collaboration

Advisor: Dr. Parastoo Abtahi

- Grounded in distributed cognition theory, analyzed the ability of LLMs to distribute complete instruction sets for complex collaborative tasks among team members, offloading cognitive burden to the environment.
- Built the CogniSplit web app and conducted a 36-participant user study in a team of 3.
- Reduced initial coordination time by 77% and task completion time by 49% without impacting task accuracy.

University of Minnesota

Jul 2024 – May 2025

Proxy Selection in Divergent Data Sets

Advisor: Dr. Howard Lavine

- Designed ProxyFinder, a transfer learning-based method that helps political science researchers identify proxy variables for unobserved constructs in survey data.
- Identified optimal proxies with 92% frequency under Monte Carlo testing, with an average of 5.5% more variance explained than state-of-the-art methods on political science datasets.

University of Minnesota

Apr 2024 – Aug 2024

The Core of Reactionary Politics

Advisor: Dr. Howard Lavine

- Studied and adapted ML & statistical methods to analyze members of American political movements who feel that their socioeconomic status is threatened.

Publications & Presentations

- **Danek K** and Ramaswamy V. *Concept-Aware Pruning via Disentangled Subspaces for Robust Convolutional Networks*. XAI4CV Workshop at CVPR, Denver, CO, June 2026. (Poster)
[\[Paper\]](#) [\[Poster\]](#) [\[Code\]](#)
- **Danek K**, Lee S, Teng H et al. *CogniSplit: LLM-Mediated Instruction Distribution for Effective Collaboration*. Preprint, 2026.
[\[Preprint\]](#) [\[Code\]](#)
- **Danek K**, Daniels J, Lavine H, and Parker C. *ProxyFinder: Methodological Proxy Selection in Divergent Data Sets*. Midwest Political Science Association (MPSA) Annual Meeting, Chicago, IL, April 2025. (Oral)
[\[Paper\]](#) [\[Code\]](#)
- **Danek K**. *ProxyFinder: Methodological Proxy Selection in Divergent Data Sets*. Princeton Research Day, Virtual, April 2025. (Oral)
[\[Recording\]](#)

Teaching

- Undergraduate Course Assistant**, Princeton University, Princeton, NJ *Sep – Dec 2024*
- COS324: Introduction to Machine Learning.
 - Graded approx. 150 students' written work and programming assignments (Python) weekly.
 - Held weekly office hours to assist students with concepts, written work, and programming in machine learning, statistics, and probability theory.

Awards

- **Sigma Xi Nomination**, Princeton University 2026
- **Cum Laude**, Princeton University 2026
- **Best Project Award**, Princeton COS436: Human-Computer Interaction 2025
CogniSplit: LLM-Mediated Collaboration
- **Project X Award**, Princeton School of Engineering and Applied Science 2025
Concept-Aware Neural Network Pruning
- **Undergraduate Fund for Academic Conferences**, Princeton University 2025
Methodological Proxy Selection in Divergent Data Sets
- **Student-Initiated Internship Program**, Office of Undergraduate Research, Princeton University 2024
Methodological Proxy Selection in Divergent Data Sets

Skills

Languages: Python, Java, C/C++, Go, Rust, R, SQL

Libraries & Tools: PyTorch, OpenCV, Hugging Face, Git/GitHub, Unix/Linux, Bash, L^AT_EX

Professional Associations

- Association for Computing Machinery – Special Interest Group on Artificial Intelligence (ACM SIGAI)
- The Computer Vision Foundation (CVF)
- Midwest Political Science Association (MPSA)